Pathology prediction from the EEG: status and diagnosis groups

📄 **PDF version**

This notebook reads the classification side of the multi-task brain-age models: how well the EEG predicts the **normal/abnormal status** and **13 NEDC diagnosis groups** of a TUH recording. For every target it reports the threshold-free ranking (ROC-AUC, PR-AUC), the **best F1** over all thresholds with its bootstrap interval, the **errors** (false positives and negatives) at that threshold, and the **ROC and precision-recall curves**.

It compares four runs on **one identical split** (Exp4mtbal's recorded `data_split.json`: 27,616 train, 3,422 validation and 274 TUAB-test recordings):

- two runs that also predict age (`Exp4mt` , `Exp4mtbal` , Table 2.10 of `brain_age_research_summary.ipynb`);
- two **classification-only** runs without the age prediction: `Exp4cls` (EEG only) and `Exp4clsage` (the recording's known age as an extra encoder input).

Per-recording probabilities come from `scripts/eval_multitask_classification.py` (`docs/data/multitask_class_predictions.csv.gz`); every metric is computed here from them by `labram/eval/binary_metrics.py` .

```
runs scored: ['Exp4mt', 'Exp4mtbal']
```

1. Runs and labels

Table 1.1. The runs

All four runs share everything except the columns below: `Exp4noearEEG` 's setup (19 channels without A1/A2/T1/T2, EEG-only input, LaBraM-base start, 8 epochs with 2 warm-up, 5 training minutes per recording per epoch, mixup, early stopping) and the multi-task head (2 shared layers 256-256, an age branch 128-64-1 and a classification branch 128-64-14). The classification loss is $0.5\,\mathrm{BCE}_{\mathrm{status}} + 0.5\,L_{\mathrm{path}}$ with L_{path} the plain (`known`) or class-balanced (`present`) mean BCE over the diagnosis groups (Eqs. (2.15)–(2.16) of the research summary). `age weight` multiplies the age (Huber) loss (`loss.age_weight`); at 0 the age branch gets no gradient and the model is a pure classifier. `age input` `tokens` (`model.age_input=tokens`) feeds the recording's known age (z-scored with the training mean 52.35 and SD 17.60 years) through a `Linear(1 → 200)` + LayerNorm as one extra token into the transformer, next to the EEG patch tokens. `selected on` is the validation metric that picks the reported epoch.

run	age weight	pathology loss	age input	selected on	where
`Exp4mt`	1	known	–	age MAE	SageMaker
`Exp4mtbal`	1	present (class-balanced)	–	age MAE	local
`Exp4cls`	**0**	present	–	pathology macro ROC-AUC	local
`Exp4clsage`	**0**	present	**tokens**	pathology macro ROC-AUC	SageMaker

Table 1.2. Labels per target

Per target: the recordings whose label is known, the positives among them and their prevalence π (Eq. (2.6)), on validation and on TUAB test. A label is unknown when the recording has no status, or for a group when its session is not in the NEDC clinical sheet; unknown labels are never scored. TUAB test holds only 11 diagnosis-group positives in total, so the groups are judged on validation and test is used for the status.

target	val known	val positives	val prevalence %	test known	test positives
status_abnormal	3064	1676	54.70	274	125
clinical_seizure	2991	185	6.19	268	0
electrographic_seizure	2991	11	0.37	268	0
epilepsy	2991	46	1.54	268	0
stroke	2991	93	3.11	268	1
tbi	2991	50	1.67	268	0
substance	2991	103	3.44	268	10
bipolar	2991	31	1.04	268	7
anxiety	2991	10	0.33	268	0
depression	2991	8	0.27	268	0
schizophrenia	2991	9	0.30	268	0
dementia	2991	10	0.33	268	1
ms	2991	7	0.23	268	0
other	2991	12	0.40	268	0

2. Metrics

For one target, recording i of a split has the label $y_i \in \{0, 1\}$ (known labels only) and the score p_i, the mean of its windows' sigmoid probabilities (as in training's per-recording metrics). Predicting positive when $p_i \geq t$ gives, at threshold t, the true and false positives and negatives $\mathrm{TP}(t), \mathrm{FP}(t), \mathrm{FN}(t), \mathrm{TN}(t)$ (counts of recordings, `binary_metrics.py::threshold_metrics`) and

$$\mathrm{prec}(t) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{rec}(t) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{spec}(t) = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{2.1}$$

$$F_1(t) = \frac{2\,\mathrm{prec}(t)\,\mathrm{rec}(t)}{\mathrm{prec}(t) + \mathrm{rec}(t)} \tag{2.2}$$

(0 when no recording is flagged). The **errors** at t are the error rate and the balanced error

$$\mathrm{err}(t) = \frac{\mathrm{FP} + \mathrm{FN}}{n}, \qquad \mathrm{berr}(t) = 1 - \tfrac{1}{2}\big(\mathrm{rec}(t) + \mathrm{spec}(t)\big) \tag{2.3}$$

with n the known recordings. The **best F1** is the maximum over every distinct score as threshold (`best_f1_threshold`; ties go to the higher threshold):

$$t^\star = \arg\max_{t \in \{p_i\}} F_1(t), \qquad F_1^\star = F_1(t^\star) \tag{2.4}$$

$F_1^\star$ on a split picks $t^\star$ on that split's own labels, so it is optimistic. The honest operating point on test carries the **validation** threshold over:

$$F_1^{\text{val}\to\text{test}} = F_1^{\text{test}}\left(t^\star_{\text{val}}\right) \tag{2.5}$$

The threshold-free metrics are the ROC-AUC (`sklearn` `roc_auc_score` , the area under $\text{rec}(t)$ against $1 - \text{spec}(t)$) and the PR-AUC as average precision (`average_precision_score`):

$$\text{AP} = \sum_k \left(\text{rec}_k - \text{rec}_{k-1}\right) \text{prec}_k, \qquad \mathbb{E}[\text{AP}_{\text{random}}] = \pi = \frac{1}{n}\sum_i y_i \tag{2.6}$$

over the thresholds k in decreasing order. A random score has AP $\approx \pi$ and ROC-AUC 0.5; flagging every recording gives $F_1 = 2\pi/(1+\pi)$, the floor marked in Figure 3.1. With 7–50 positives per rare group, every metric carries a **95 % stratified bootstrap interval** (`bootstrap_ci` , $B = 1000$, positives and negatives resampled separately, $t^\star$ re-chosen in each resample):

$$\left[\, q_{0.025}\{m^{(b)}\}_{b=1}^{B},\ q_{0.975}\{m^{(b)}\}_{b=1}^{B} \,\right] \tag{2.7}$$

with $m^{(b)}$ the metric on resample b and q the empirical quantile. The macro value of a metric is its mean over the 13 groups (status excluded):

$$m_{\text{macro}} = \frac{1}{13}\sum_{k=1}^{13} m_k \tag{2.8}$$

3. Results

Table 3.1. Ranking quality on validation: ROC-AUC and PR-AUC

Per target and run, on the 3,422 validation recordings (known labels only, Table 1.2): ROC-AUC and PR-AUC (Eq. (2.6)) with their 95 % bootstrap intervals (Eq. (2.7)) as `value [low, high]` ; `prevalence` is the PR-AUC of a random score. The last row is the macro mean over the 13 groups (Eq. (2.8)).

target	prevalence	ROC-AUC Exp4mt	ROC-AUC Exp4mtbal	PR-AUC Exp4mt	PR-AUC Exp4mtbal
status_abnormal	0.547	0.822 [0.806, 0.836]	0.820 [0.804, 0.836]	0.821 [0.802, 0.840]	0.819 [0.799, 0.838]
clinical_seizure	0.062	0.778 [0.750, 0.805]	0.789 [0.760, 0.819]	0.180 [0.148, 0.227]	0.204 [0.166, 0.256]
electrographic_seizure	0.004	0.963 [0.933, 0.985]	0.962 [0.947, 0.975]	0.083 [0.047, 0.199]	0.051 [0.039, 0.083]
epilepsy	0.015	0.740 [0.692, 0.786]	0.680 [0.620, 0.739]	0.030 [0.025, 0.041]	0.024 [0.020, 0.032]
stroke	0.031	0.756 [0.706, 0.800]	0.769 [0.720, 0.813]	0.083 [0.067, 0.116]	0.105 [0.077, 0.164]
tbi	0.017	0.764 [0.713, 0.813]	0.742 [0.681, 0.799]	0.046 [0.033, 0.086]	0.052 [0.034, 0.111]
substance	0.034	0.618 [0.557, 0.674]	0.611 [0.552, 0.666]	0.057 [0.045, 0.078]	0.058 [0.044, 0.088]
bipolar	0.010	0.556 [0.471, 0.641]	0.532 [0.448, 0.624]	0.012 [0.010, 0.016]	0.011 [0.009, 0.016]
anxiety	0.003	0.740 [0.554, 0.895]	0.770 [0.597, 0.914]	0.031 [0.007, 0.145]	0.035 [0.008, 0.165]
depression	0.003	0.633 [0.372, 0.865]	0.641 [0.421, 0.851]	0.011 [0.004, 0.040]	0.009 [0.003, 0.042]
schizophrenia	0.003	0.551 [0.487, 0.622]	0.703 [0.628, 0.774]	0.003 [0.003, 0.005]	0.005 [0.005, 0.008]
dementia	0.003	0.848 [0.779, 0.910]	0.890 [0.851, 0.928]	0.015 [0.010, 0.045]	0.025 [0.013, 0.105]
ms	0.002	0.522 [0.364, 0.682]	0.642 [0.424, 0.806]	0.003 [0.002, 0.007]	0.004 [0.003, 0.009]
other	0.004	0.623 [0.493, 0.735]	0.652 [0.539, 0.761]	0.006 [0.005, 0.009]	0.006 [0.005, 0.010]
macro (13 groups)	–	0.699	0.722	0.043	0.045

Table 3.2. Best F1 and the errors at its threshold, validation

Per target and run: the best F1 $F_1^\star$ (Eq. (2.4)) with its 95 % interval (Eq. (2.7)), the threshold $t^\star$ (a probability), precision and recall at it (Eq. (2.1)), the errors as recording counts — false positives FP and false negatives FN next to the true positives TP — and the error rate and balanced error (Eq. (2.3)) in percent. The last row per run is the macro mean of $F_1^\star$ (Eq. (2.8)).

run	target	best F1	t*	precision	recall	TP	FP	FN	error %	balanced error %
Exp4mt	**status_abnormal**	0.799 [0.789, 0.812]	0.43	0.75	0.86	1436	481	240	23.53	24.49
	clinical_seizure	0.247 [0.225, 0.288]	0.124	0.17	0.46	85	418	100	17.32	34.48
	electrographic_seizure	0.182 [0.114, 0.342]	0.098	0.11	0.46	5	39	6	1.50	27.93
	epilepsy	0.067 [0.058, 0.100]	0.0267	0.04	0.67	31	843	15	28.69	30.62
	stroke	0.162 [0.142, 0.206]	0.0528	0.09	0.59	55	530	38	18.99	29.57
	tbi	0.107 [0.071, 0.187]	0.0225	0.06	0.34	17	252	33	9.53	37.28
	substance	0.132 [0.096, 0.187]	0.0778	0.11	0.17	18	151	85	7.89	43.88
	bipolar	0.028 [0.023, 0.042]	0.0109	0.01	0.74	23	1615	8	54.26	40.18
	anxiety	0.148 [0.018, 0.345]	0.0253	0.12	0.20	2	15	8	0.77	40.25
	depression	0.049 [0.014, 0.121]	0.0137	0.03	0.25	2	71	6	2.57	38.69
	schizophrenia	0.010 [0.010, 0.012]	0.00145	0.01	1.00	9	1800	0	60.18	30.18
	dementia	0.050 [0.025, 0.158]	0.0226	0.03	0.10	1	29	9	1.27	45.49
	ms	0.008 [0.006, 0.024]	0.0036	0.00	0.14	1	242	6	8.29	46.91
	other	0.016 [0.012, 0.024]	0.000802	0.01	0.83	10	1265	2	42.36	29.57
	macro (13 groups)	0.093	–	0.06	0.46	–	–	–	19.51	36.54
Exp4mtbal	**status_abnormal**	0.795 [0.785, 0.808]	0.416	0.74	0.85	1432	493	244	24.05	25.04
	clinical_seizure	0.270 [0.236, 0.319]	0.159	0.20	0.43	79	321	106	14.28	34.37
	electrographic_seizure	0.125 [0.086, 0.205]	0.0766	0.07	0.64	7	94	4	3.28	19.76
	epilepsy	0.062 [0.048, 0.085]	0.0409	0.03	0.43	20	584	26	20.39	38.18
	stroke	0.163 [0.145, 0.232]	0.0806	0.10	0.37	34	291	59	11.70	36.74
	tbi	0.095 [0.072, 0.206]	0.0439	0.08	0.12	6	70	44	3.81	45.19
	substance	0.127 [0.091, 0.196]	0.0656	0.09	0.21	22	221	81	10.10	43.15
	bipolar	0.025 [0.023, 0.044]	0.018	0.01	0.45	14	1070	17	36.34	45.49
	anxiety	0.143 [0.024, 0.353]	0.0678	0.11	0.20	2	16	8	0.80	40.27
	depression	0.043 [0.009, 0.138]	0.043	0.03	0.12	1	38	7	1.50	44.39
	schizophrenia	0.015 [0.012, 0.020]	0.0108	0.01	0.78	7	902	2	30.22	26.23
	dementia	0.100 [0.031, 0.300]	0.102	0.10	0.10	1	9	9	0.60	45.15
	ms	0.012 [0.007, 0.025]	0.0122	0.01	0.57	4	644	3	21.63	32.22
	other	0.016 [0.012, 0.026]	0.0126	0.01	0.50	6	710	6	23.94	36.92
	macro (13 groups)	0.092	–	0.07	0.38	–	–	–	13.74	37.54

Table 3.3. TUAB test: best F1 and the validation threshold carried over

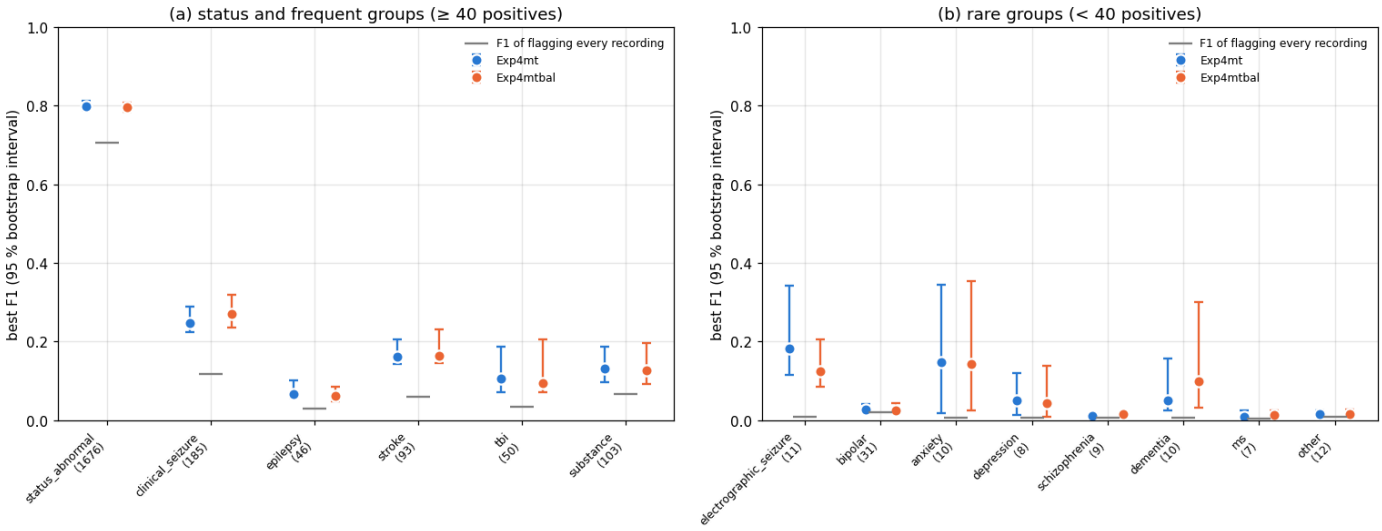
The 274 TUAB-test recordings, for the targets with test positives: ROC-AUC, the best F1 on test $F_1^\star$ (Eq. (2.4), optimistic: its threshold is chosen on test), and the honest $F_1^{\mathrm{val}\to\mathrm{test}}$ (Eq. (2.5))

with its errors (FP, FN) and balanced error (Eq. (2.3)). Only the status has enough test positives (125) to read; the groups have 1–10.

target	run	positives	ROC-AUC	best F1 (test t*)	F1 at val t*	val t*	TP	FP	FN	balanced error % at val t*
status_abnormal	Exp4mt	125	0.92	0.85	0.82	0.43	111	36	14	17.68
	Exp4mtbal	125	0.92	0.84	0.82	0.416	111	34	14	17.01
stroke	Exp4mt	1	0.73	0.03	0.00	0.0528	0	36	1	56.74
	Exp4mtbal	1	0.76	0.03	0.00	0.0806	0	19	1	53.56
substance	Exp4mt	10	0.51	0.14	0.12	0.0778	1	5	9	45.97
	Exp4mtbal	10	0.53	0.12	0.00	0.0656	0	16	10	53.10
bipolar	Exp4mt	7	0.72	0.13	0.07	0.0109	6	161	1	37.99
	Exp4mtbal	7	0.71	0.20	0.08	0.018	5	119	2	37.08
dementia	Exp4mt	1	0.90	0.07	0.00	0.0226	0	2	1	50.38
	Exp4mtbal	1	0.93	0.09	0.00	0.102	0	2	1	50.38

Figure 3.1. Best F1 per target with its bootstrap interval, validation

$F_1^\star$ (Eq. (2.4)) per target and run with its 95 % interval (Eq. (2.7)) on the 3,422 validation recordings; the number under each target is its positives. (a) Status and the groups with at least 40 positives; (b) the rare groups. The grey bar is the F1 of flagging every recording, $2\pi/(1+\pi)$ (Eq. (2.6)): a model must clear it to be of any use as a detector.



Figures 3.2–3.15. ROC and precision-recall curves per target

One figure per target (status first, then the 13 groups), on validation (3,422 recordings, known labels only); each run is one line in its fixed colour, the dot is its best-F1 operating point $t^\star$ (Eq. (2.4)). (a) ROC curve: recall against the false-positive rate $1-\mathrm{spec}$ (Eq. (2.1)), chance on the diagonal; legend: ROC-AUC. (b) Precision-recall curve; chance is the prevalence π (Eq. (2.6)), the grey contours are constant F1 (Eq. (2.2)); legend: average precision and $F_1^\star$. The status figure (3.2) is followed by its TUAB-test version (3.2(test), 274 recordings).

Figure 3.2. status_abnormal: ROC and precision-recall curves, validation

1676 positives of 3064 recordings (prevalence 0.547).

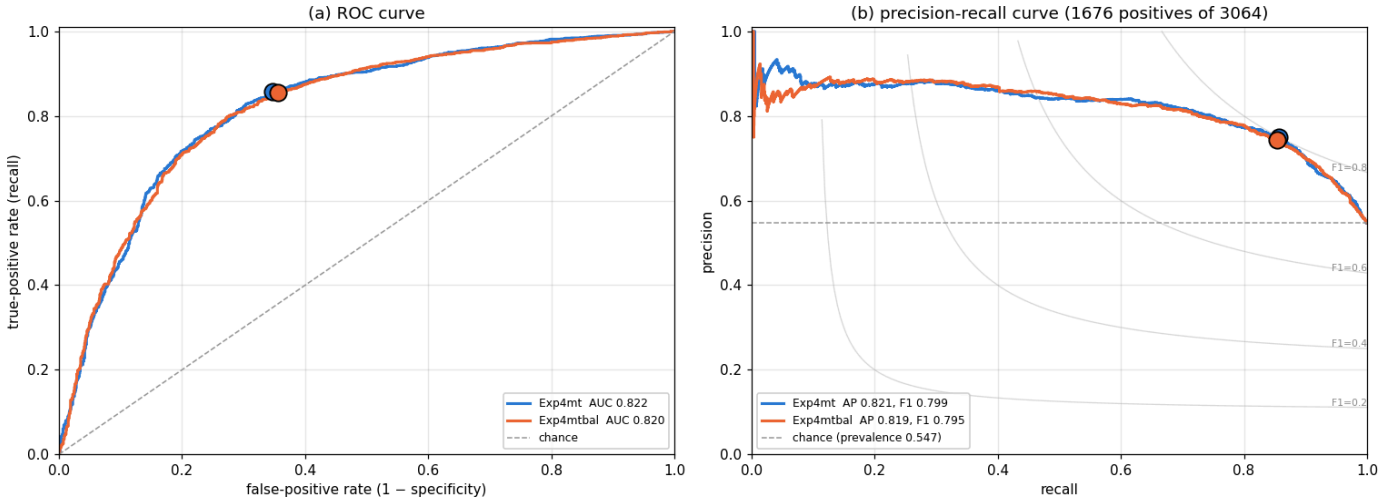


Figure 3.2(test). status_abnormal: ROC and precision-recall curves, TUAB test

274 TUAB-test recordings, 125 abnormal.

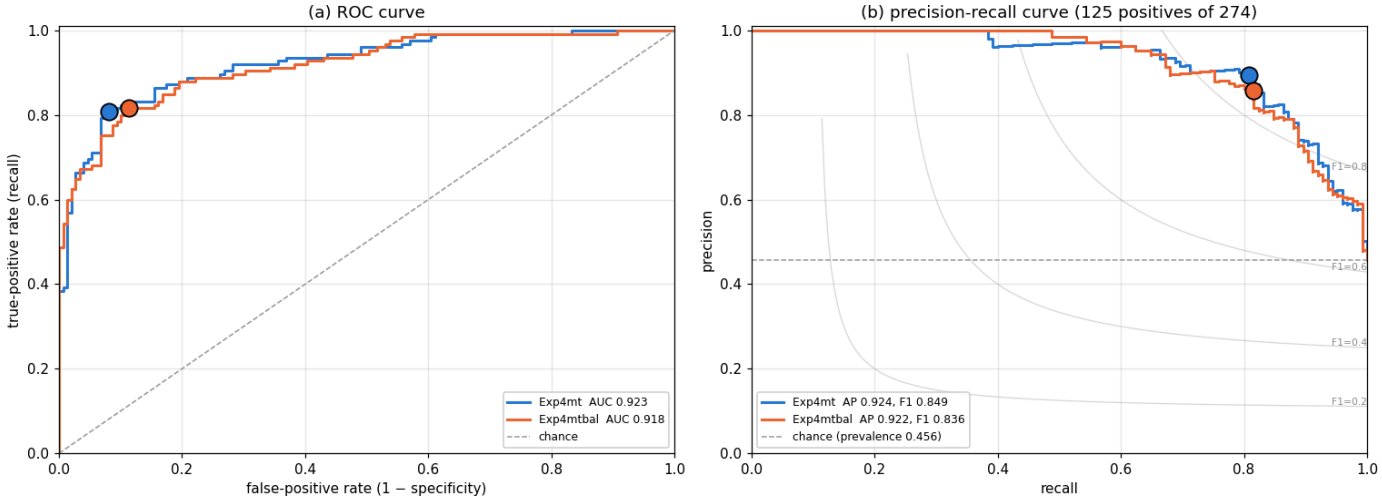


Figure 3.3. clinical_seizure: ROC and precision-recall curves, validation

185 positives of 2991 recordings (prevalence 0.062).

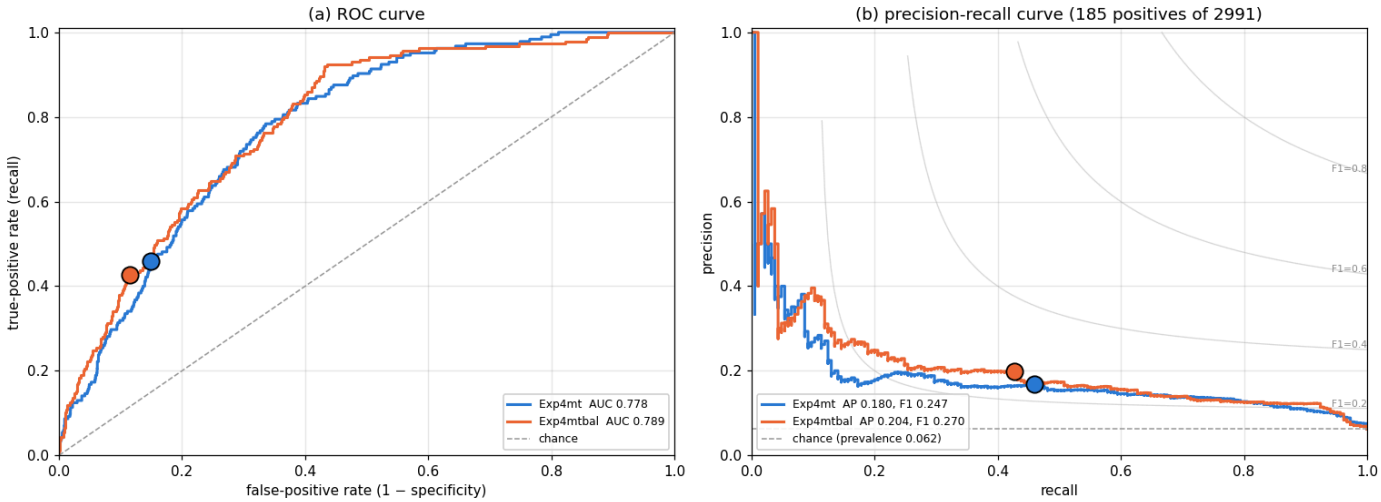


Figure 3.4. electrographic_seizure: ROC and precision-recall curves, validation

11 positives of 2991 recordings (prevalence 0.004).

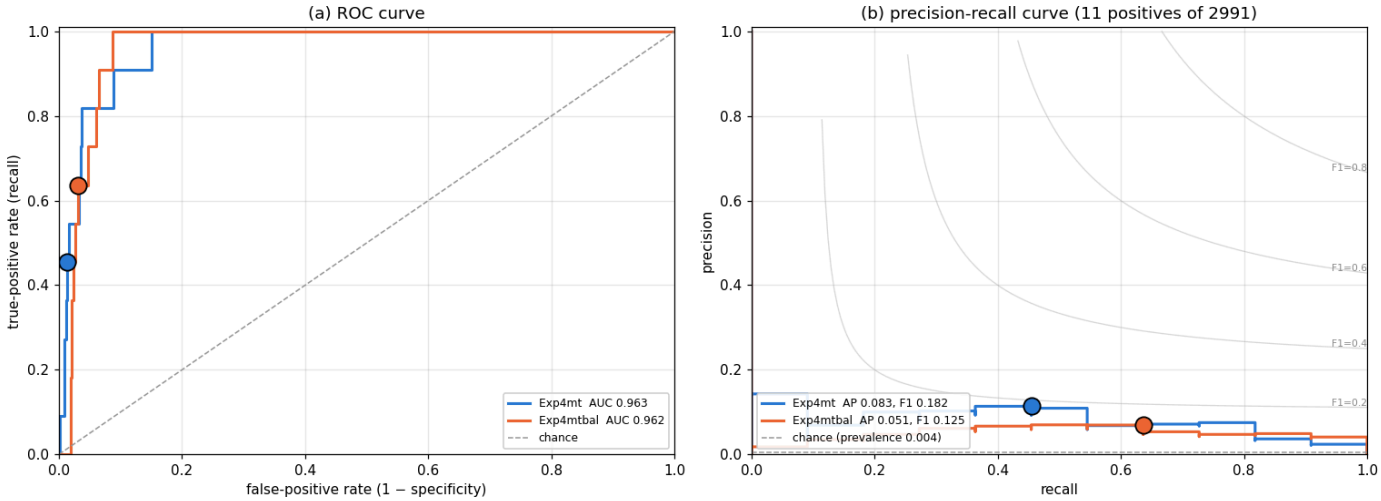


Figure 3.5. epilepsy: ROC and precision-recall curves, validation

46 positives of 2991 recordings (prevalence 0.015).

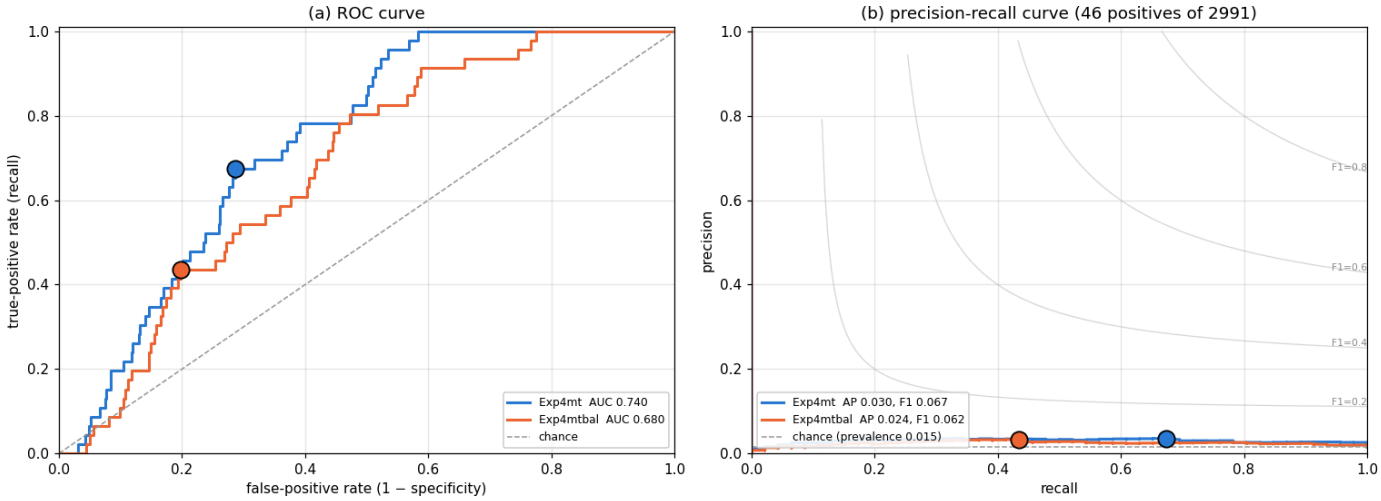


Figure 3.6. stroke: ROC and precision-recall curves, validation

93 positives of 2991 recordings (prevalence 0.031).

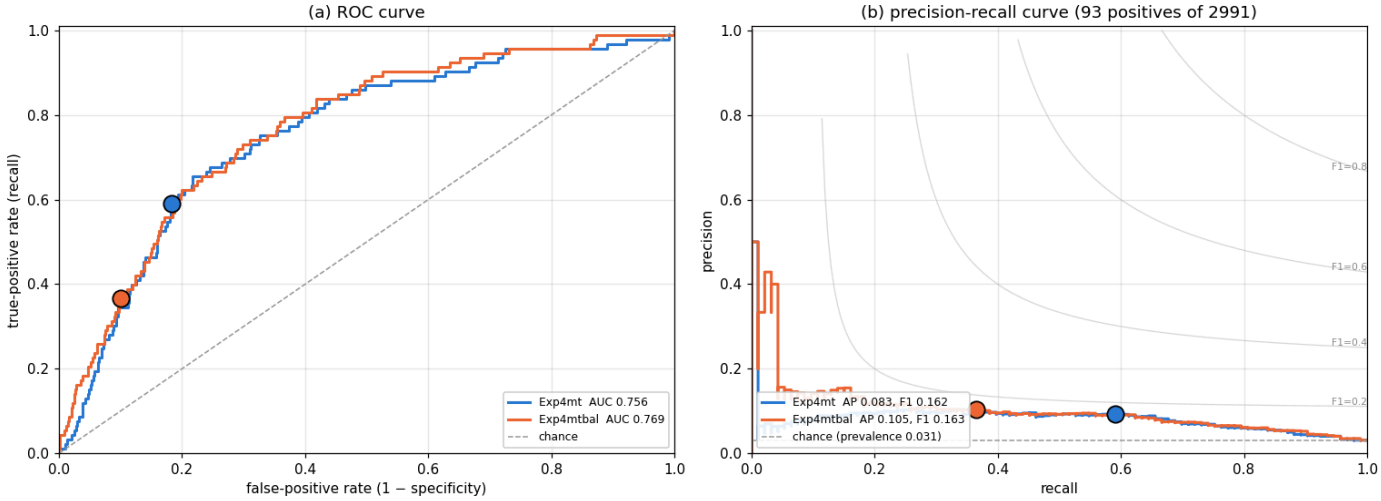


Figure 3.7. tbi: ROC and precision-recall curves, validation

50 positives of 2991 recordings (prevalence 0.017).

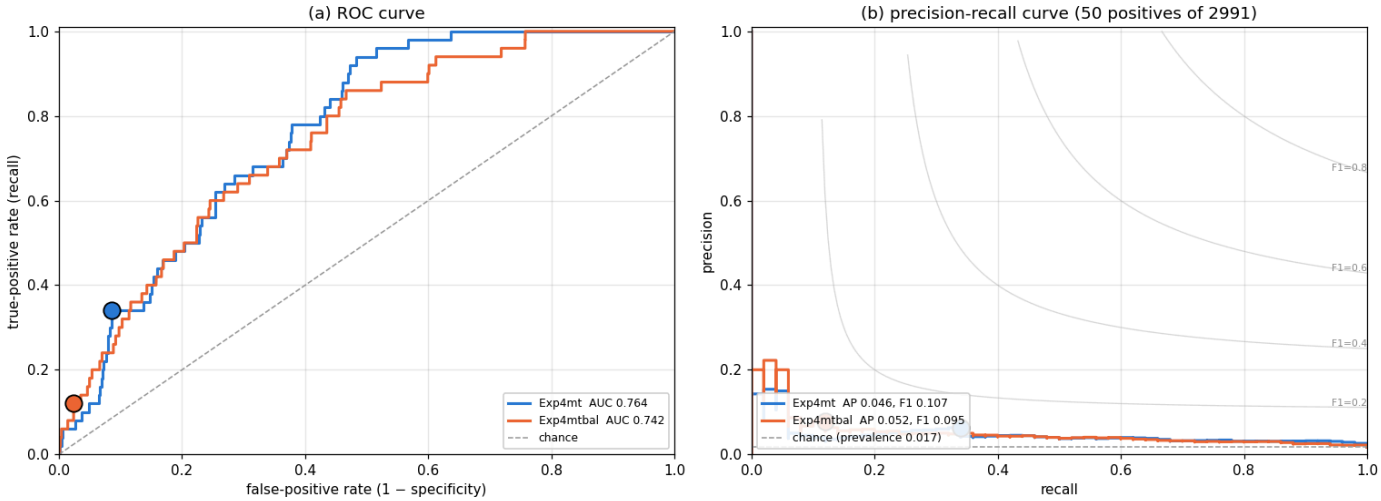


Figure 3.8. substance: ROC and precision-recall curves, validation

103 positives of 2991 recordings (prevalence 0.034).

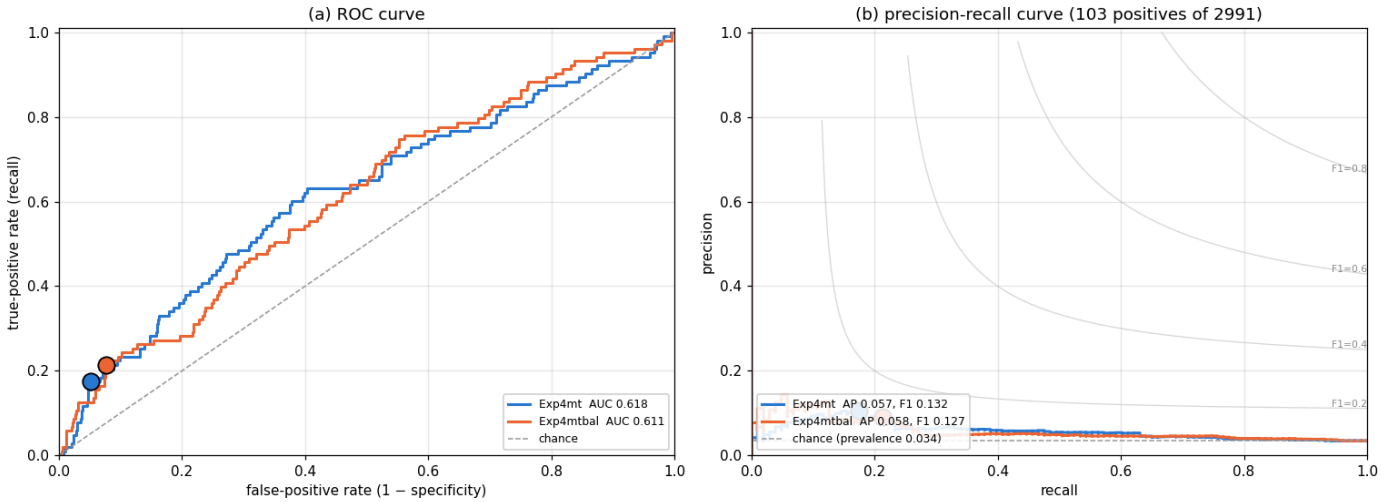


Figure 3.9. bipolar: ROC and precision-recall curves, validation

31 positives of 2991 recordings (prevalence 0.010).

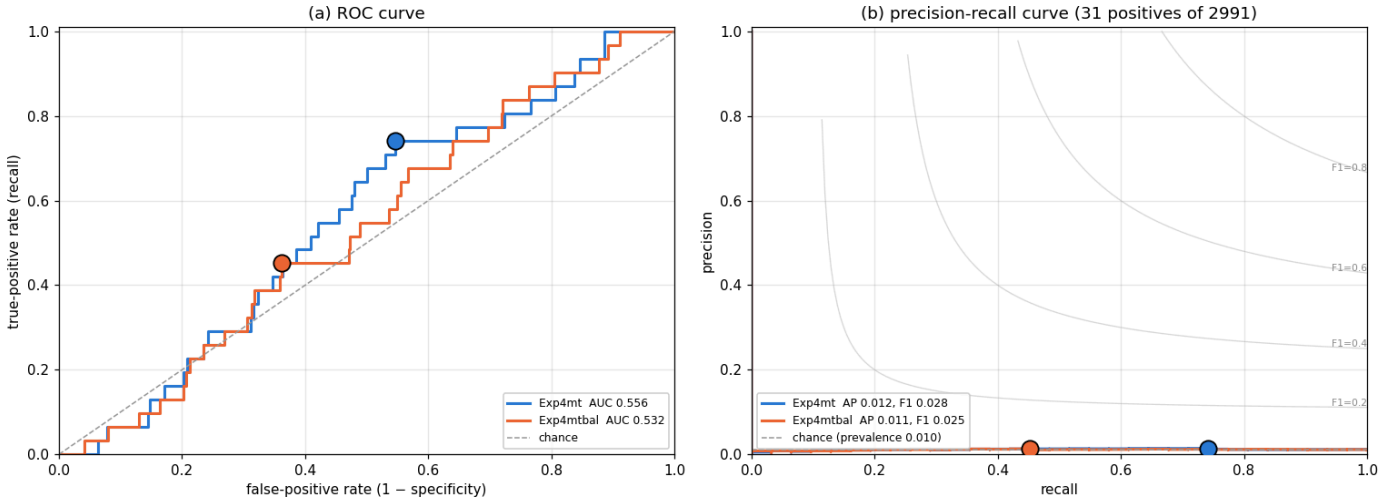


Figure 3.10. anxiety: ROC and precision-recall curves, validation

10 positives of 2991 recordings (prevalence 0.003).

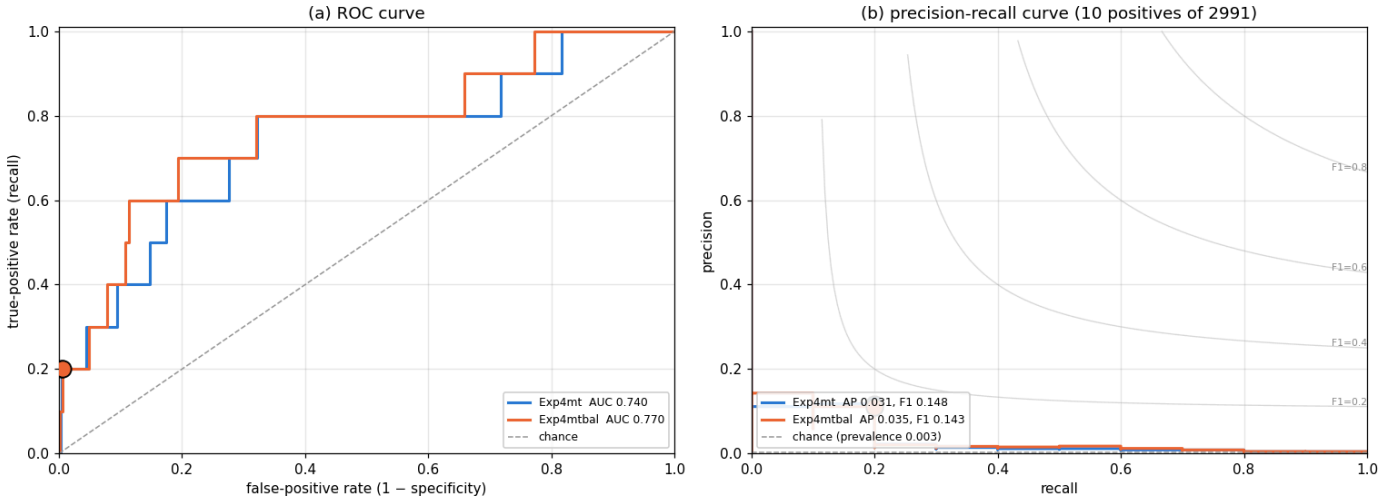


Figure 3.11. depression: ROC and precision-recall curves, validation

8 positives of 2991 recordings (prevalence 0.003).

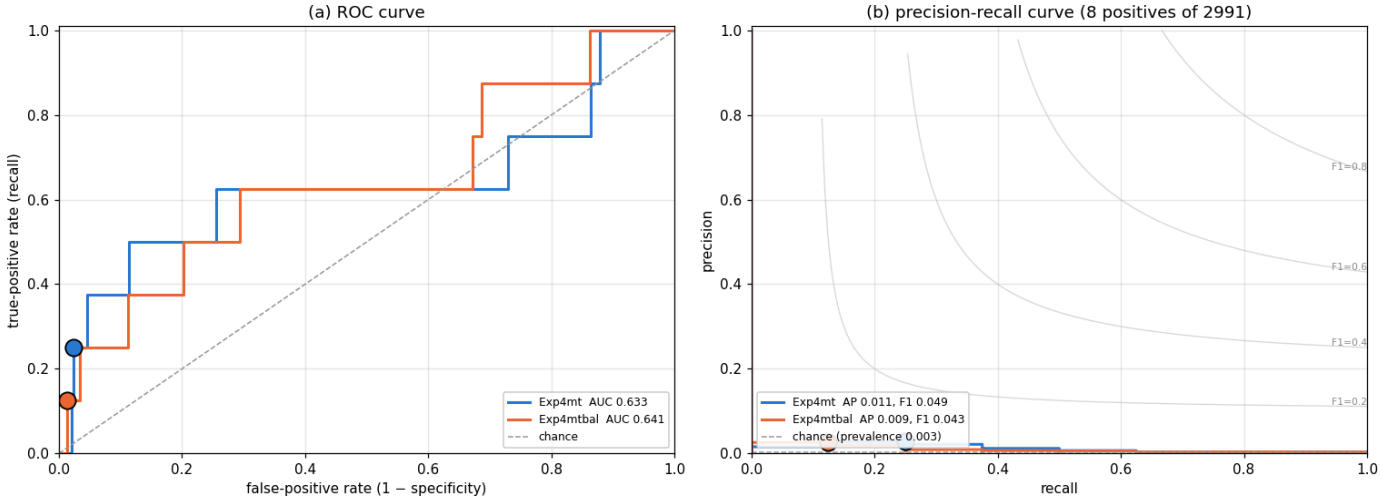


Figure 3.12. schizophrenia: ROC and precision-recall curves, validation

9 positives of 2991 recordings (prevalence 0.003).

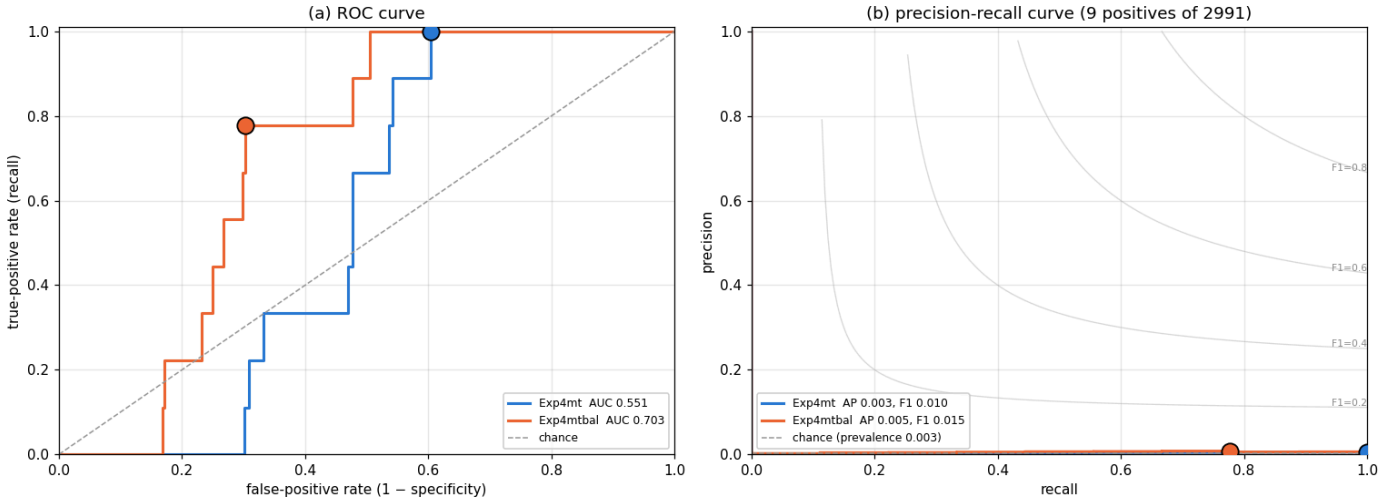


Figure 3.13. dementia: ROC and precision-recall curves, validation

10 positives of 2991 recordings (prevalence 0.003).

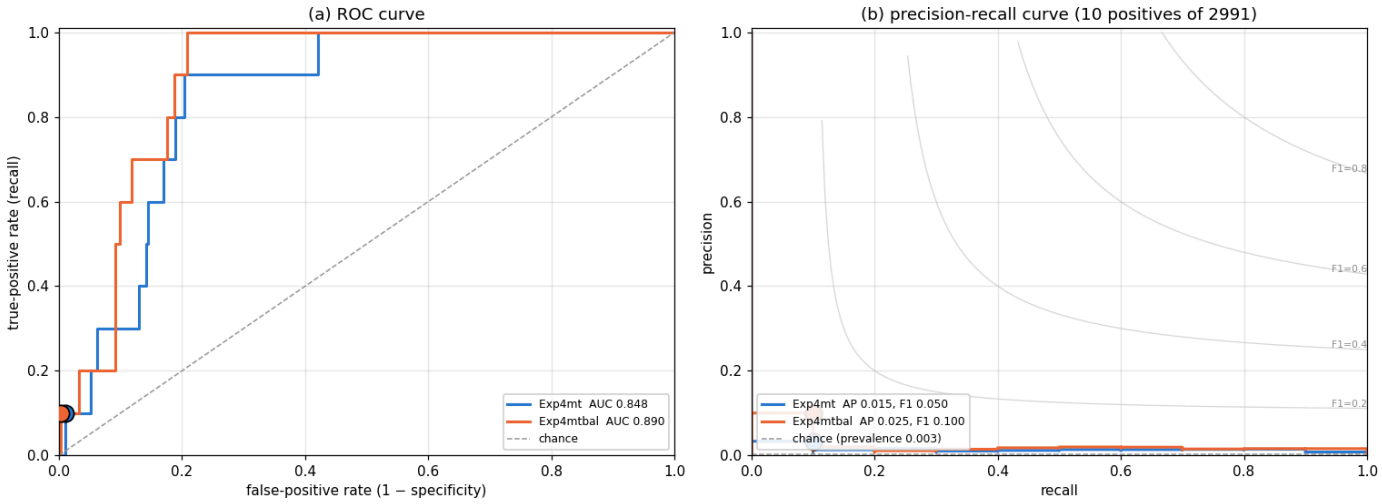


Figure 3.14. ms: ROC and precision-recall curves, validation

7 positives of 2991 recordings (prevalence 0.002).

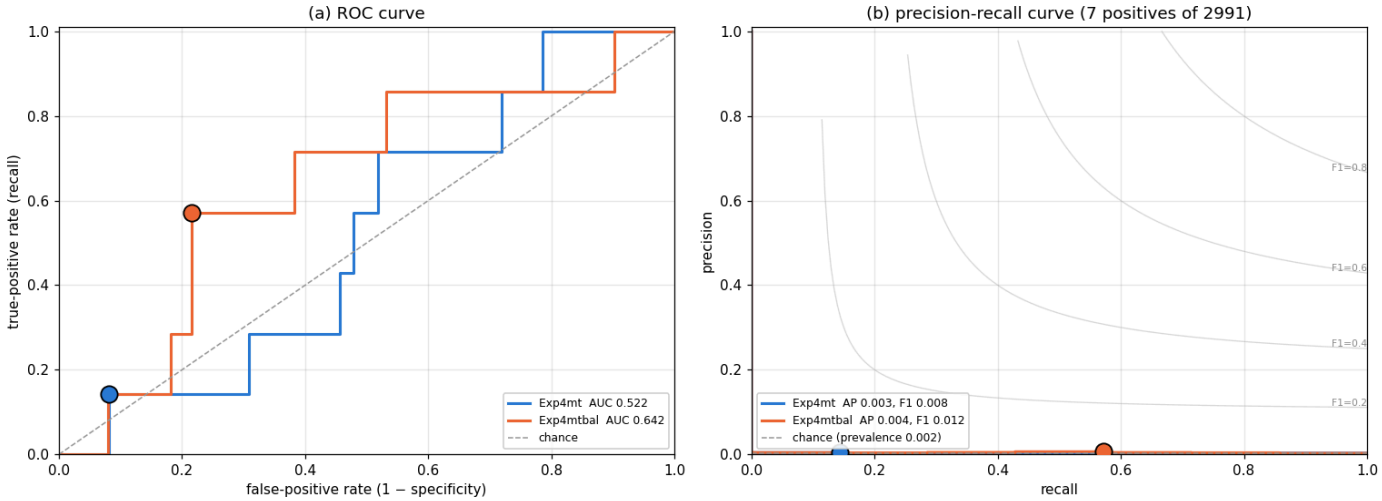
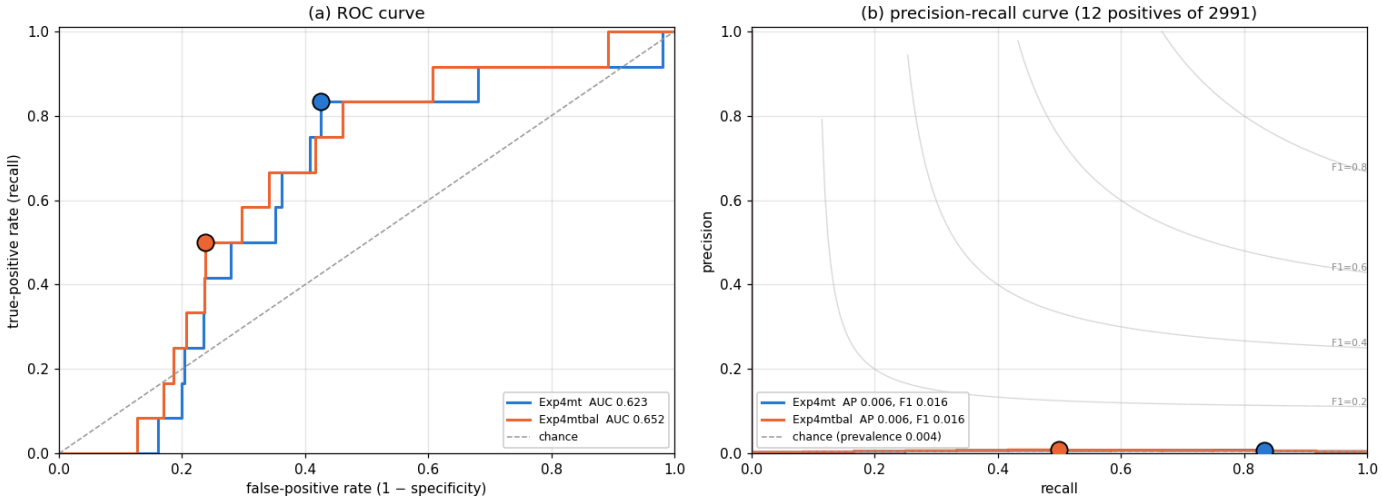


Figure 3.15. other: ROC and precision-recall curves, validation

12 positives of 2991 recordings (prevalence 0.004).



4. Reading the results

Table 4.1. Classification-only runs against the age-predicting run with the same loss

Each later run minus `Exp4mtbal` (same class-balanced loss, same split), on validation: ROC-AUC, PR-AUC and $F_1^\star$ of the status, and their macro means over the 13 groups (Eq. (2.8)). Positive = better than `Exp4mtbal`. For a single rare group a difference below about 0.1 ROC-AUC is inside its bootstrap interval (Table 3.1); the macro values average that noise down.

The classification-only runs are not scored yet.

Reading Tables 3.1–3.3 and Figures 3.1–3.15.

- **Status is a usable detector.** On validation the best F1 is 0.80 (Table 3.2) at a threshold near 0.43, with precision 0.75 and recall 0.86; at that point `Exp4mt` makes 481 false positives and 240 false negatives out of 3,064 recordings (error 23.5 %). On TUAB test the validation threshold carries over well (Table 3.3): F1 0.82 at the validation $t^\star$ against 0.84–0.85 when $t^\star$ is chosen on test, with 111 of 125 abnormal recordings found, 34–36 false alarms and 14 misses (balanced error 17–18 %). Figure 3.2(test) shows the curves.
- **Clinical seizure and stroke are above the flag-everyone floor but far from a detector.** Clinical seizure reaches F1 0.25–0.27 against a floor of 0.12 (Figure 3.1(a)). Its best operating point flags about 400–500 recordings to find 80 of the 185 seizure recordings: about 4–5 false alarms per hit (Table 3.2). Stroke reaches 0.16 against 0.06. In Figure 3.3(b) precision stays below 0.4 at every recall.
- **A high ROC-AUC does not make a rare group detectable.** Electrographic seizure has ROC-AUC 0.96 (Table 3.1), yet its best F1 is 0.13–0.18, with an interval reaching 0.34. At prevalence 0.004, `Exp4mt`'s best point finds 5 of the 11 positives with a false-positive rate of only 1.3 %, but that is still 39 false alarms. Dementia (ROC-AUC 0.85–0.89) is similar.
- **Many groups sit at the floor.** For bipolar, schizophrenia, MS and "other", $t^\star$ falls to about 0.01: the "best" F1 is then reached by flagging most recordings (e.g. schizophrenia: 1,800 false positives for 9 positives), and it is barely above the flag-everyone floor (Figure 3.1(b)). These groups are not predicted.
- **The class-balanced loss changes the ranking, not the detection.** `Exp4mtbal` raises the macro ROC-AUC from 0.70 to 0.72 (Table 3.1), but the macro best F1 stays at 0.09 (Table 3.2). Its gains are on groups too rare to give a usable F1.